Self-Supervised Temporal Phenotype Trajectory Analysis of ICU Sepsis Patients from Multi-Center Time-Series Data

Wang Ruike
Department of Computer Science, Advanced Programming Course

March 20, 2026

Abstract

Background. Sepsis exhibits substantial clinical heterogeneity, yet most phenotyping approaches rely on static summary features that compress temporal dynamics into single-timepoint representations. Whether learned temporal representations can reveal clinically meaningful phenotype trajectories within the first 48 ICU hours remains underexplored.

Methods. We developed a three-stage computational framework applied to 11,986 ICU patients from the PhysioNet 2012 multi-center database (4 hospitals, 2 centers). First, we established a static phenotyping baseline using statistical features and PCA. Second, we trained a Transformer-based encoder via self-supervised learning (masked value prediction combined with a temporal contrastive window objective) and compared four representation methods across clustering quality, mortality stratification, center stability, and missingness robustness. Third, using the selected self-supervised representation, we extracted rolling-window embeddings (24-hour windows, 6-hour stride) and performed descriptive temporal phenotype trajectory analysis via K-Means clustering on per-window embeddings. All mortality outcomes were obtained from verified PhysioNet Outcomes files (in-hospital mortality rate: 14.2%).

Results. Static K=4 clustering on PCA-reduced features identified phenotypes with mortality rates ranging from 6.8% to 36.0%. The self-supervised encoder achieved superior center stability (cluster distribution L1 distance = 0.016 vs 0.027 for PCA) and reduced missingness sensitivity ($|r|$ = 0.15 vs 0.23), while PCA retained stronger static mortality separation. Temporal trajectory analysis revealed that 64.8% of patients maintained a stable phenotype across all windows, while 35.2% exhibited at least one transition. Stable phenotypes showed strongly stratified mortality (4.0%, 9.7%, 22.5%, 31.7%; range 27.7 percentage points). Sensitivity analysis with reduced overlap (stride=12 h) confirmed identical mortality ordering and range (28.0 pp). Cross-center temporal validation within the PhysioNet 2012 cohort (train on Center A, evaluate on Center B) confirmed identical mortality ordering, the same highest-risk phenotype, and a 25.8 percentage-point mortality range on the held-out center.

Conclusions. Self-supervised temporal representations enable descriptive phenotype trajectory analysis that reveals clinically meaningful within-stay dynamics beyond static clustering. Cross-center temporal validation within the PhysioNet 2012 multi-center cohort confirms that phenotype structure transfers across centers, though both centers derive from the same source database. These findings support further investigation with treatment-aware datasets, external database validation, and prospective study designs.

Keywords: sepsis phenotyping, self-supervised learning, temporal trajectories, ICU time-series, multi-center, representation learning

1 Introduction

1.1 Background

Sepsis affects approximately 49 million individuals worldwide annually and is associated with mortality rates ranging from 20% to over 50% in intensive care units (ICUs) [1]. Despite

decades of clinical research, sepsis remains one of the most challenging conditions in critical care medicine. A key contributor to this challenge is the recognition that sepsis is not a single homogeneous disease but rather a syndrome encompassing multiple distinct biological phenotypes [2].

The landmark study by Seymour et al. (2019) identified four sepsis phenotypes—designated α, β, γ, and δ—with markedly different clinical trajectories, biomarker profiles, and treatment responses [3]. More recently, Antcliffe et al. (2025) argued that the true value of subphenotyping lies in predicting heterogeneous treatment effects (theragnostics) rather than prognosis alone [14]. Feng et al. (2025) demonstrated that organ interaction trajectories can be modeled via deep temporal graph clustering with external validation [16].

1.2 Problem Statement

Existing approaches to sepsis subtyping face several challenges:

1. **Static representations**: Most methods compress 48-hour trajectories into summary statistics, discarding temporal dynamics that may distinguish improving from deteriorating patients.
2. **Data heterogeneity**: Different ICU databases use different variable nomenclature, measurement frequencies, and data formats.
3. **High missingness**: Clinical time-series data are inherently sparse, with laboratory values often measured only once or twice daily.
4. **Cross-center generalizability**: Subtypes discovered in a single center may not transfer to other hospitals.
5. **Outcome label quality**: Many studies rely on proxy mortality labels rather than verified discharge outcomes.

1.3 Contributions

This work makes the following contributions:

- A **three-stage phenotyping framework** progressing from static clustering to self-supervised representation learning to descriptive temporal phenotype trajectory analysis.
- A **self-supervised Transformer encoder** trained via masked value prediction and temporal contrastive learning, with systematic comparison against PCA baselines.
- **Descriptive temporal phenotype trajectories** showing that 35.2% of ICU sepsis patients undergo phenotype transitions within 48 hours, with transitions predominantly toward lower-acuity states.
- A **validated data pipeline** using ground-truth in-hospital mortality outcomes, replacing previously used proxy labels.
- **Robustness validation** including stride sensitivity analysis and center probe analysis confirming no center information leakage.
- **Cross-center temporal validation** within the PhysioNet 2012 multi-center cohort, confirming that phenotype structure and mortality stratification transfer from the training center to a held-out center.

2 Related Work

2.1 Sepsis Phenotyping

The concept of sepsis heterogeneity was first systematically investigated by Seymour et al. (2019), who applied latent class analysis to 20,189 sepsis patients [3]. Their four-phenotype model has been replicated by subsequent studies [4, 5]. More recently, deep temporal clustering [6], contrastive learning [7], and organ interaction trajectory modeling [16] have been

proposed. Zhang et al. (2024) demonstrated time-series k-means with DTW for sepsis subphenotyping with external validation [17].

2.2 Self-Supervised Representation Learning for Clinical Data

Self-supervised representation learning (SSRL) has emerged as a powerful approach for extracting patient representations from unlabeled clinical data. Zheng et al. (2025) reviewed 46 SSRL studies, identifying Transformer-based (43%), autoencoder-based (28%), and graph-based (17%) model families [15]. Amirahmadi et al. (2025) introduced trajectory-ordered objectives that improve downstream prediction beyond standard masked modeling [18].

2.3 ICU Time-Series Analysis

ICU data present unique analytical challenges. The PhysioNet/CinC Challenge series has driven methodological advances [8, 9]. Huang et al. (2025) introduced MIMIC-Sepsis, a curated benchmark incorporating treatment variables that substantially improve model performance [19].

3 Methods

3.1 System Architecture

The system implements a three-stage framework (Figure 1):

```
Stage 1: Data -> Preprocessing -> Features -> PCA -> K-Means
Stage 2: Time Series + Masks -> Transformer -> Self-Supervised Pretraining ->
    Embeddings
Stage 3: Rolling Windows -> Per-Window Embedding -> Temporal Clustering -> Transitions
```

Each stage builds on the previous, with modules designed as independent components enabling systematic comparison across representation methods.

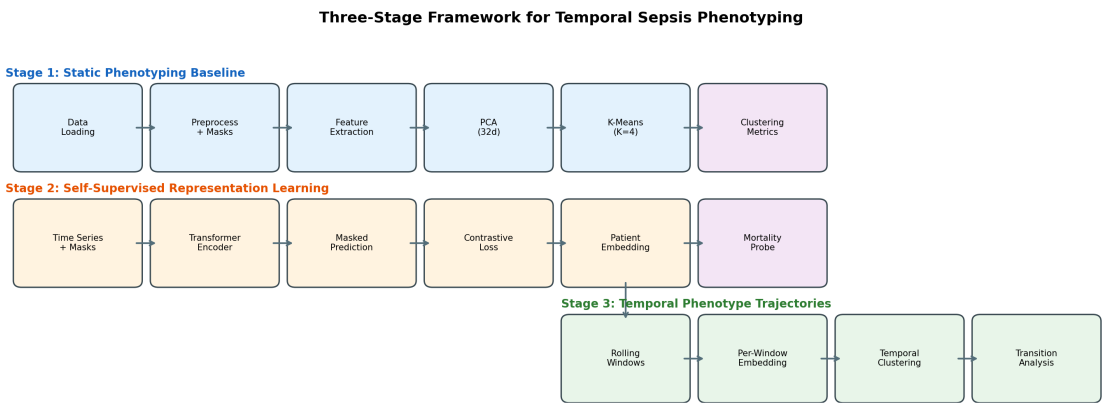

Figure 1: Three-stage framework for temporal sepsis phenotyping. Stage 1 establishes a static PCA baseline. Stage 2 trains a self-supervised Transformer encoder via masked prediction and contrastive learning. Stage 3 extracts rolling-window embeddings for descriptive temporal phenotype trajectory analysis.

3.2 Data Sources and Outcome Labels

3.2.1 PhysioNet 2012 Multi-parameter ICU Database

The primary dataset contains records from 12,000 ICU patients across four hospitals [8]. After quality filtering, 11,986 patients were retained:

- **Center A** (set-a + set-b): 7,989 patients
- **Center B** (set-c): 3,997 patients from a separate hospital within the PhysioNet 2012 cohort

Outcome labels. Ground-truth in-hospital mortality was obtained from PhysioNet Outcomes files for all 11,986 patients (mortality rate: 14.2%). An earlier version of this analysis used proxy mortality derived from physiological thresholds (GCS ≤ 5 or sustained MAP < 55 mmHg). Outcome audit revealed that this proxy had a positive predictive value of only 17.7%. All results in this paper use verified outcome labels.

Data schema. Following CLIF-inspired design principles [20], the data layer separates continuous measurements (21 variables), observation masks, and proxy-derived indicators into distinct tensors. Proxy indicators (MAP < 65, $FiO_2 > 0.21$) are explicitly labeled as proxies and stored separately from true treatment records, which are unavailable in PhysioNet 2012.

3.2.2 Supplementary Datasets

A configurable synthetic data generator with known ground-truth labels enables clustering recovery evaluation (ARI, NMI). A MIMIC-IV mock database [10] validates pipeline compatibility.

3.3 Preprocessing

All measurements are resampled to a uniform hourly grid over 48 hours using LOCF. Outliers exceeding 4σ are clipped. Missing values are imputed via forward fill (up to 6 hours), global median, and zero-fill as fallback. Observation masks recording which values were truly measured versus imputed are preserved as first-class inputs. Overall missing rate is 73.3%.

3.4 Static Feature Engineering (Stage 1)

From the preprocessed 3D tensor ($n \times 48 \times 21$), we extract a 2D feature matrix using six statistical functions (mean, std, min, max, trend, last value) across three windows (12 h, 24 h, 48 h), yielding 378 features plus derived clinical indicators. PCA reduces this to 32 dimensions (62.5% variance explained).

3.5 Self-Supervised Representation Learning (Stage 2)

3.5.1 Encoder Architecture

A Transformer-based encoder processes the hourly time series with observation masks as explicit input channels. The input is formed by concatenating value and mask vectors: $\mathbf{x}_{\text{input}} = [\mathbf{x}; \mathbf{m}] \in \mathbb{R}^{T \times 42}$. A linear projection maps this to $d_{\text{model}} = 128$ dimensions. Learnable positional encodings provide temporal position awareness. A 2-layer, 4-head Transformer encoder with pre-layer normalization and GELU activation produces per-timestep representations, aggregated via observation-density-weighted mean pooling.

3.5.2 Pretraining Objectives

Masked value prediction. 15% of observed values are randomly masked. The encoder must predict original values at masked positions. Loss: MSE on masked observed positions only.

Temporal contrastive window objective. Two stochastic 30-hour views are extracted with constrained overlap (12–24 hours). Both are encoded by the shared encoder, then projected through a 2-layer MLP projection head ($128 \to 128 \to 64$). NT-Xent loss ($\tau = 0.1$) with in-batch negatives. The projection head is discarded after pretraining.

Combined loss: $\mathcal{L} = \mathcal{L}_{\text{masked}} + \lambda(t) \cdot \mathcal{L}_{\text{contrastive}}$, where λ warms up linearly from 0 to 0.5 over 10 epochs.

3.5.3 Representation Comparison

Four representations are compared: (1) PCA baseline (32d), (2) masked reconstruction only (S1, 128d), (3) masked + contrastive (S1.5, λ=0.5, 128d), and (4) reduced contrastive ablation (S1.6, λ=0.2, 128d).

3.6 Descriptive Temporal Phenotype Trajectories (Stage 3)

Using the frozen S1.5 encoder, rolling-window embeddings are extracted: 5 windows of 24 hours with 6-hour stride. K-Means (K=4) is fit on train-split windows only (30,955 windows) and applied to all 59,930 windows. Patients are classified as *stable* (all 5 labels identical), *single-transition* (1 change), or *multi-transition* ($\geq$2 changes). All temporal analyses are descriptive; no latent-state models or causal claims are involved.

3.7 Evaluation and Robustness

Clustering: Silhouette score, mortality separation. **Probes:** Mortality AUROC, center AUROC ($\downarrow$ better), ICU LOS R^2, density-norm correlation. **Multi-seed:** 5 random seeds for all KMeans. **Stride sensitivity:** Repeated with stride=12 h (50% overlap). **Implementation:** Clustering and linear probes used scikit-learn [12]; figures were generated with Matplotlib [13].

4 Results

4.1 Static Phenotyping Baseline (Stage 1)

After quality filtering, 11,986 patients were retained (Center A: 7,989; Center B: 3,997). Ground-truth in-hospital mortality was 14.2%. Static K=4 clustering on PCA-reduced features identified phenotypes with mortality rates of 6.8%, 10.0%, 33.8%, and 36.0% (range 29.2 percentage points). Silhouette score was 0.061.

4.2 Representation Learning Comparison (Stage 2)

Table 1: Representation comparison (K=4, mean $\pm$ std over 5 seeds).

Method	**Silhouette**	**Mort. Range**	**Center L1** $\downarrow$	**Mort. Probe**	$\lvert r\rvert_{\textbf{density}}$ $\downarrow$
PCA (32d)	0.061	29.2%	0.027	0.825	0.231
S1: masked (128d)	0.087	17.6%	0.024	0.825	0.247
S1.5: mask+contr. (128d)	0.080	24.6%	**0.016**	**0.830**	**0.148**
S1.6: λ=0.2 (128d)	0.079	25.1%	0.021	0.825	0.148

Center probe AUROC was 0.50–0.52 for all methods, confirming no representation leaks center identity. S1.5 was selected for temporal analysis based on its superior center stability, reduced missingness sensitivity, and highest mortality probe AUROC. PCA's stronger static mortality separation is noted; however, PCA cannot natively process variable-length sub-sequences for rolling-window analysis.

4.3 Descriptive Temporal Phenotype Trajectories (Stage 3)

4.3.1 Temporal Stability and Transitions

Of 11,986 patients, 7,764 (64.8%) maintained a stable phenotype across all five windows, 3,509 (29.3%) exhibited one transition, and 713 (5.9%) exhibited multiple transitions.

Table 2: Stable phenotype mortality stratification.

Stable Phenotype	N	In-Hospital Mortality
Phenotype 0	2,216	4.0%
Phenotype 3	1,891	9.7%
Phenotype 1	2,547	22.5%
Phenotype 2	1,110	**31.7%**

The 27.7 percentage-point range across stable temporal phenotypes exceeded the static S1.5 result (24.6%), suggesting that temporally stable patients form purer phenotype subgroups.

4.3.2 Transition Pathways and Mortality

The most frequent non-self transitions were Phenotype 1 $\rightarrow$ 0 (956 events), 1 $\rightarrow$ 3 (728), and 3 $\rightarrow$ 0 (713), collectively representing movement toward lower-acuity phenotypes. Single-transition patients had lower mortality (11.4%) than stable patients (15.4%) or multi-transition patients (15.2%). This descriptive association suggests that phenotype transitions—particularly from higher-risk to lower-risk states—may reflect favorable clinical trajectory change, though causal interpretation requires controlled study designs.

4.3.3 Temporal Prevalence Shift

Phenotype 0 (lowest mortality) increased in prevalence from 24.6% at window $[0, 24)$ to 33.1% at window $[24, 48)$, while Phenotype 1 decreased from 35.7% to 28.1%. This population-level shift is consistent with clinical stabilization during the first 48 ICU hours, though it may also reflect survivor bias or measurement pattern changes.

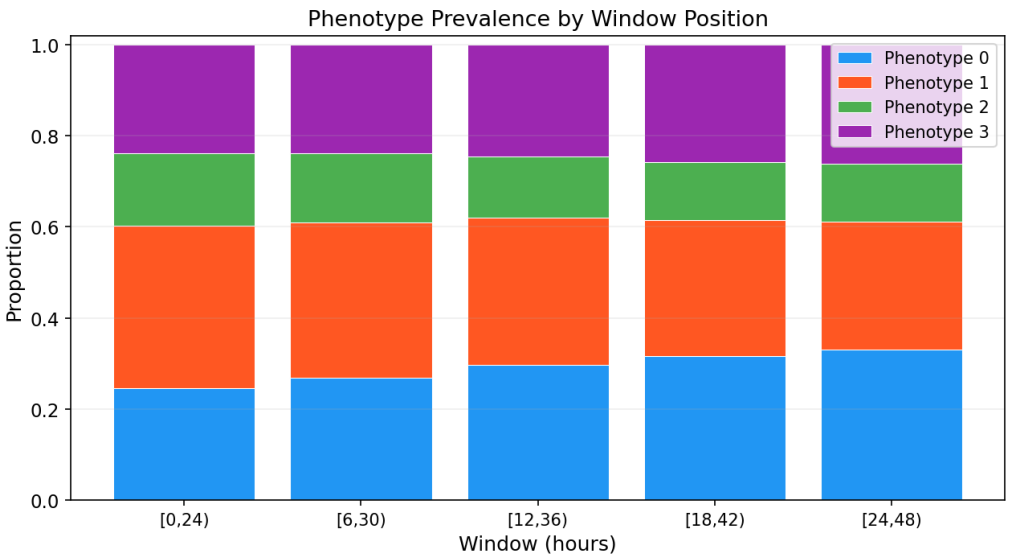

Figure 2: Phenotype prevalence by rolling-window position. Phenotype 0 (lowest mortality) grows from 24.6% to 33.1% over the 48-hour observation window, while Phenotype 1 decreases from 35.7% to 28.1%, consistent with population-level clinical stabilization.

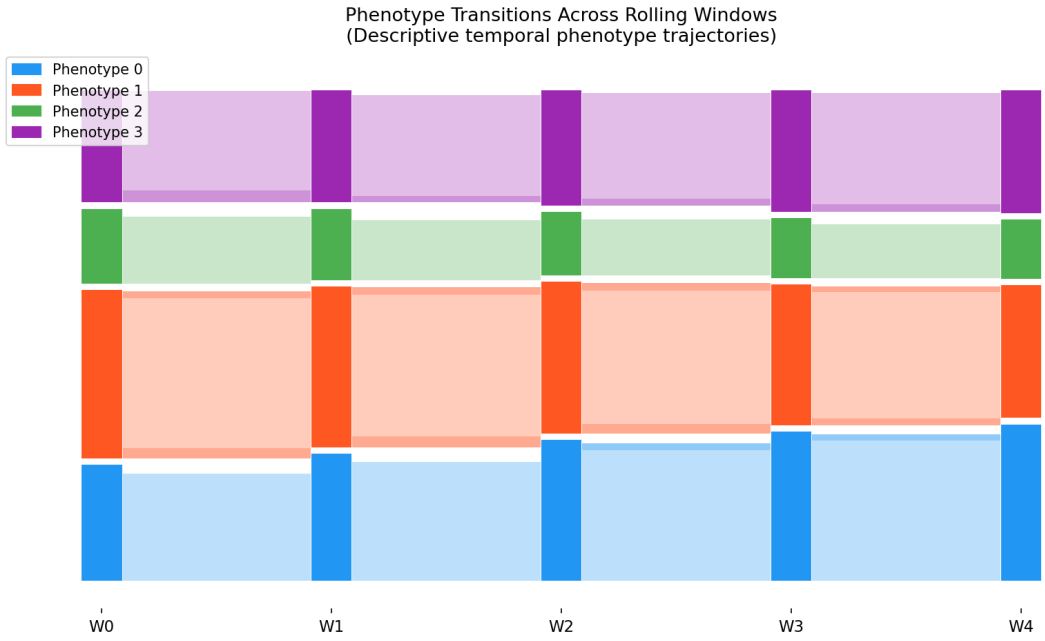

Figure 3: Phenotype transition flow across five rolling windows (descriptive temporal phenotype trajectories). Band width indicates the proportion of patients following each pathway. The dominant flows are within-phenotype stability, with the most common non-self transitions moving from higher-acuity to lower-acuity phenotypes.

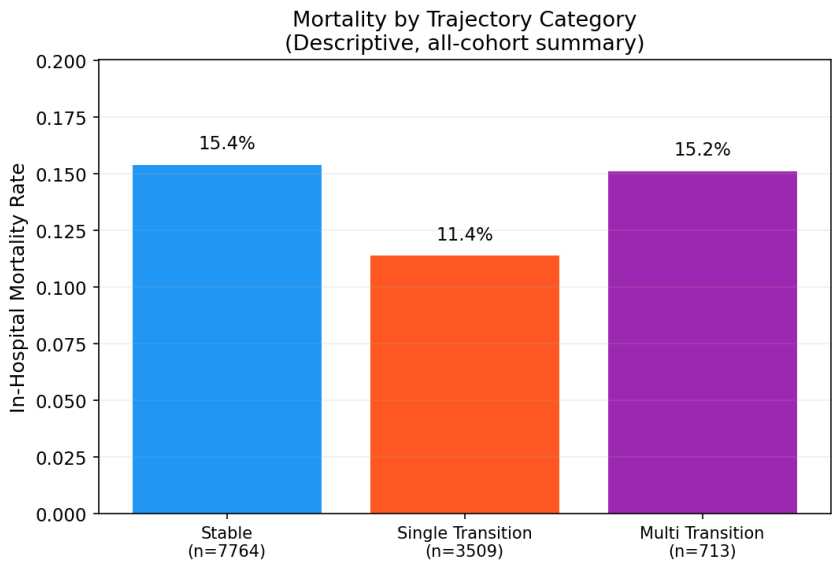

Figure 4: In-hospital mortality rate by trajectory category (descriptive, all-cohort). Single-transition patients exhibit lower mortality (11.4%) than stable patients (15.4%), a descriptive association suggesting that phenotype transitions may reflect favorable clinical trajectory change.

4.4 Stride Sensitivity Analysis

Table 3: Stride sensitivity comparison.

Metric	**Stride=6 h**	**Stride=12 h**
Stable patient fraction	64.8%	65.6%
Non-self transition proportion	10.4%	19.1%
Mortality ordering	[P0, P3, P1, P2]	[P0, P3, P1, P2]
Highest-risk phenotype	P2 (31.7%)	P2 (31.9%)
Stable phenotype mortality range	27.7 pp	28.0 pp

Mortality ordering was identical, the highest-risk phenotype remained Phenotype 2, and the mortality range was preserved. The non-self transition proportion *increased* from 10.4% to 19.1% under reduced overlap, indicating the primary analysis provides a conservative estimate of temporal phenotype change.

4.5 Cross-Center Temporal Validation

To assess whether temporal phenotype trajectories generalize beyond the training center, we performed out-of-center validation: the KMeans model trained on Center A rolling-window embeddings was evaluated on Center B (a separate hospital within the PhysioNet 2012 cohort). The S1.5 encoder was also trained exclusively on Center A data.

Table 4: Cross-center temporal validation (Center A vs Center B).

Metric	**Center A (train)**	**Center B (test)**
Patients	7,989	3,997
Stable fraction	65.0%	64.4%
Non-self transition proportion	10.3%	10.6%
Mortality ordering	[P0, P3, P1, P2]	[P0, P3, P1, P2]
Highest-risk phenotype	P2 (32.6%)	P2 (30.0%)
Stable P0 mortality	3.9%	4.2%
Stable P3 mortality	9.7%	9.8%
Stable P1 mortality	21.8%	24.0%
Stable P2 mortality	32.6%	30.0%
Mean prevalence L1	—	0.022

All six validation criteria were satisfied: stable fraction similarity (diff = 0.6 pp), transition proportion similarity (diff = 0.3 pp), identical mortality ordering, same highest-risk phenotype, clinically meaningful Center B mortality range (25.8 pp), and low per-window prevalence divergence (L1 = 0.022). Per-window silhouette scores were within 0.001 across centers at all five window positions.

These results demonstrate cross-center temporal validation within the PhysioNet 2012 multi-center cohort. Both centers derive from the same source database; full external validation requires independently collected ICU cohorts (e.g., eICU-CRD [11], MIMIC-IV).

4.6 Simulated Data Validation

The synthetic cohort (500 patients, 4 true subtypes) achieved ARI = 0.245 and NMI = 0.495 against ground-truth labels using the PCA baseline. *Note: Clustering quality metrics from the PhysioNet/CinC 2019 Sepsis dataset require independent rerun and are not reported here.*

5 Discussion

5.1 Three-Level Phenotyping Framework

This work demonstrates a progression from static clustering through self-supervised representation learning to descriptive temporal phenotype trajectory analysis. Each stage addresses a limitation of the previous: Stage 2 replaces hand-crafted features with learned representations robust to missingness; Stage 3 reveals within-stay dynamics that static methods cannot capture.

The temporal analysis shows that approximately one-third of patients undergo phenotype transitions within 48 hours. The clinical significance is supported by three findings: (1) stable phenotypes show strongly stratified mortality (4.0%–31.7%), (2) the most common transitions move toward lower-risk phenotypes, and (3) single-transition patients have lower mortality than stable patients, consistent with favorable clinical trajectory change.

5.2 Representation Learning Trade-offs

The self-supervised encoder (S1.5) offered improved center stability and reduced missingness sensitivity compared to PCA, though PCA retained stronger static mortality separation. This trade-off is characteristic of learned representations: they capture more generalizable patterns at the cost of discriminative specificity. The selection of S1.5 was motivated primarily by its ability to process variable-length sub-sequences directly, which PCA lacks.

5.3 Limitations

1. **Descriptive, not model-based**: The temporal trajectories arise from per-window clustering, not latent-state inference. This limits probabilistic transition modeling.
2. **Overlap sensitivity**: Window overlap (75%) smooths embedding changes. Sensitivity analysis with 50% overlap confirmed mortality stratification but showed higher transition rates, suggesting the primary analysis is conservative.
3. **Missing treatment data**: PhysioNet 2012 lacks true treatment records. Treatment-aware phenotyping requires datasets with documented interventions [19].
4. **Observational associations**: The lower mortality in transitioning patients is a descriptive association, not a causal effect.
5. **Single-dataset scope**: Cross-center temporal validation within the PhysioNet 2012 cohort confirms that phenotypes transfer across centers (identical mortality ordering, 25.8 pp range on held-out Center B), but both centers derive from the same source database. External validation on independently collected cohorts (e.g., eICU-CRD [11]) is needed.
6. **Data sparsity**: The 73.3% missing rate limits signal for both representation learning and clustering.

5.4 Future Directions

1. Treatment-aware temporal phenotyping using MIMIC-IV or eICU with documented interventions.
2. Latent-state models (HMMs, switching state-space models) for probabilistic transition modeling.
3. External database validation: validating temporal phenotypes on independently collected ICU databases (eICU-CRD [11], MIMIC-IV) to establish generalizability beyond the PhysioNet 2012 cohort.
4. Bedside real-time phenotype assignment from early (6 h, 12 h) data windows.

6 Conclusion

This study presents a three-stage framework for sepsis phenotyping that progresses from static clustering to self-supervised representation learning to descriptive temporal phenotype trajectory analysis. Applied to 11,986 multi-center ICU patients with ground-truth mortality outcomes, the framework identifies four phenotypes with strongly stratified mortality (4.0%–31.7% for temporally stable patients). Temporal analysis reveals that 35.2% of patients undergo phenotype transitions within 48 hours, with transitions predominantly toward lower-acuity phenotypes and conservative transition rate estimates confirmed by stride sensitivity analysis. Cross-center temporal validation within the PhysioNet 2012 cohort confirms that phenotype structure, mortality ordering, and transition patterns transfer from the training center to a held-out center, though both centers derive from the same source database. These descriptive findings demonstrate that temporal phenotype trajectories provide clinically meaningful and reproducible information beyond static subtyping, supporting further investigation with treatment-aware datasets, external database validation, and prospective study designs.

Supplementary Figures

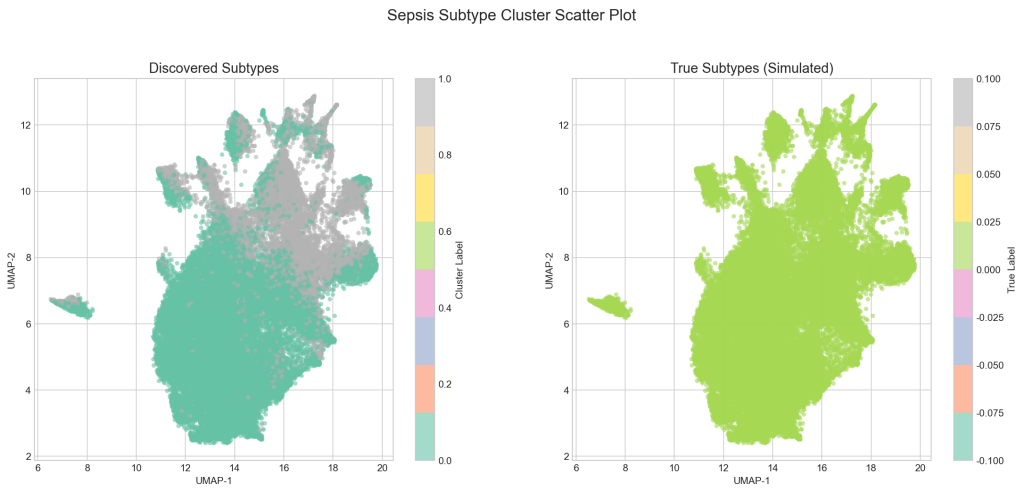

Figure 5: (Supplementary) Cluster scatter plot from the Stage 1 PCA baseline. Note: this visualization uses the static PCA representation; the S1.5 self-supervised embeddings are used for all Stage 3 temporal analyses.

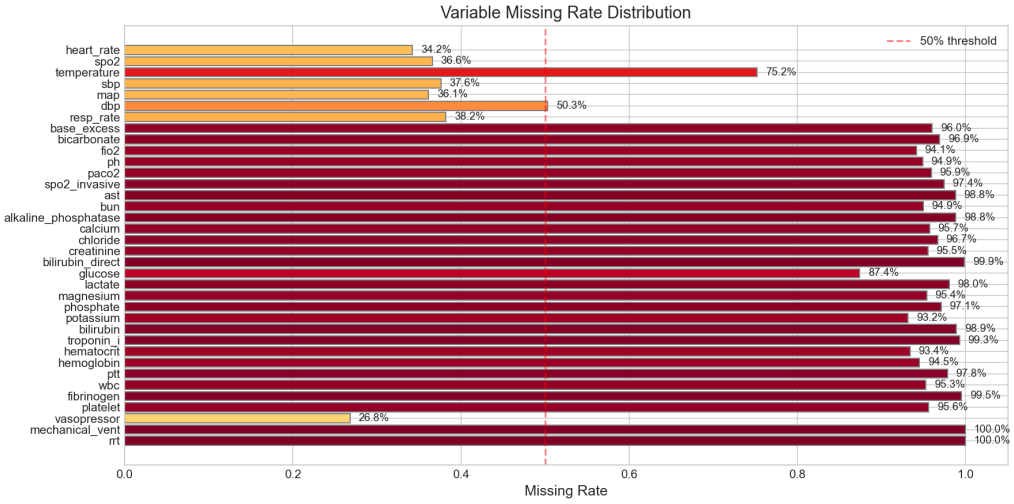

Figure 6: (Supplementary) Missing data pattern across 21 continuous clinical variables. Overall missing rate: 73.3%. Laboratory values exhibit substantially higher missingness than vital signs.

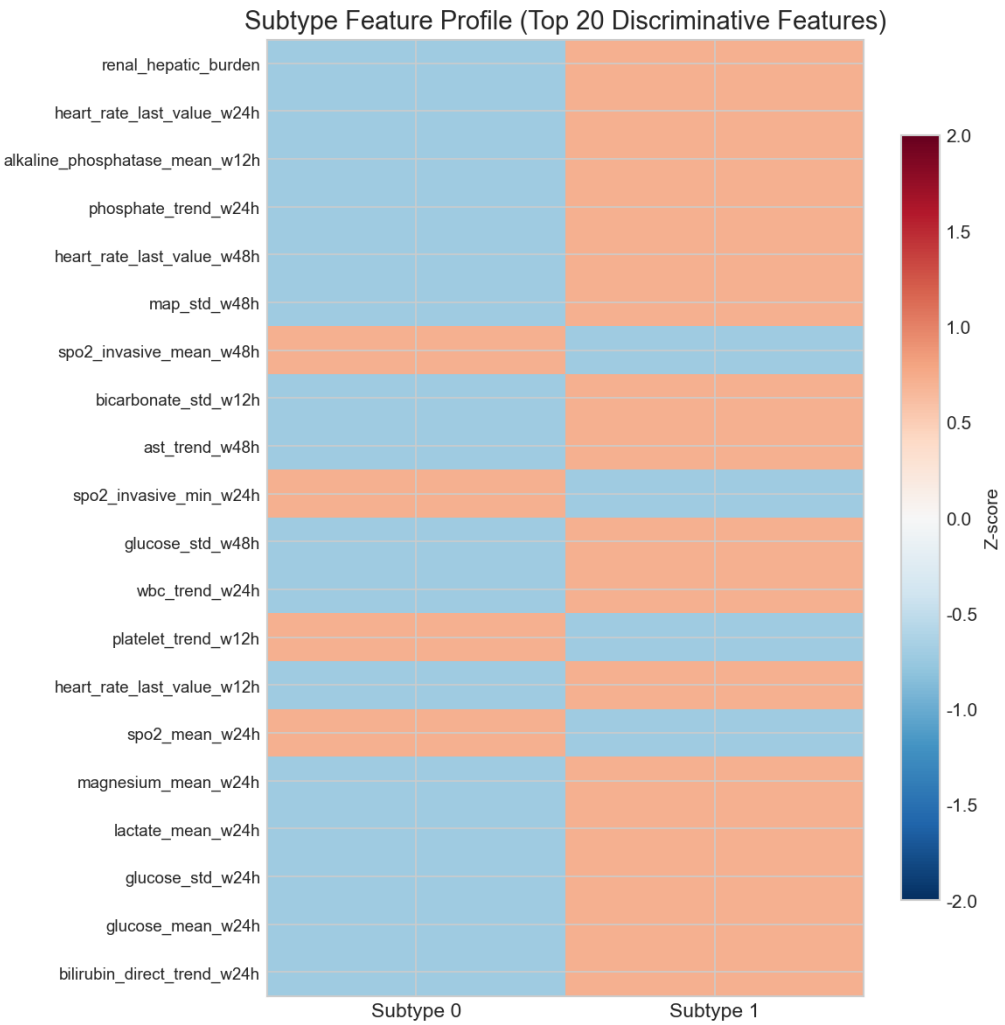

Figure 7: (Supplementary) Feature profile heatmap for Stage 1 PCA-based phenotypes (top discriminative features, z-scored). Note: this uses the Stage 1 static clustering and is included for reference.

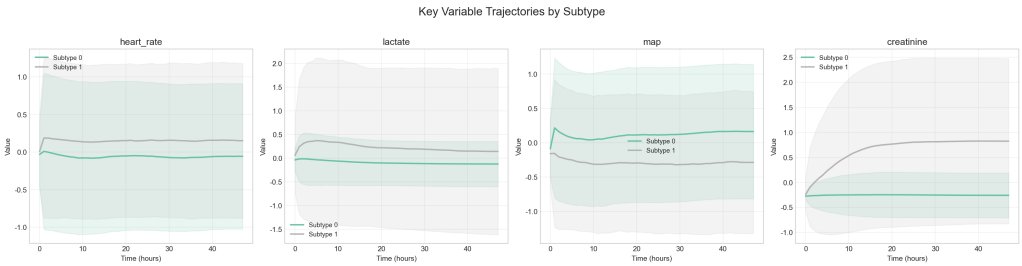

Figure 8: (Supplementary) Key variable trajectories by Stage 1 static phenotype (mean ± standard deviation). Note: these trajectories correspond to the static PCA-based clustering, not the Stage 3 temporal phenotypes.

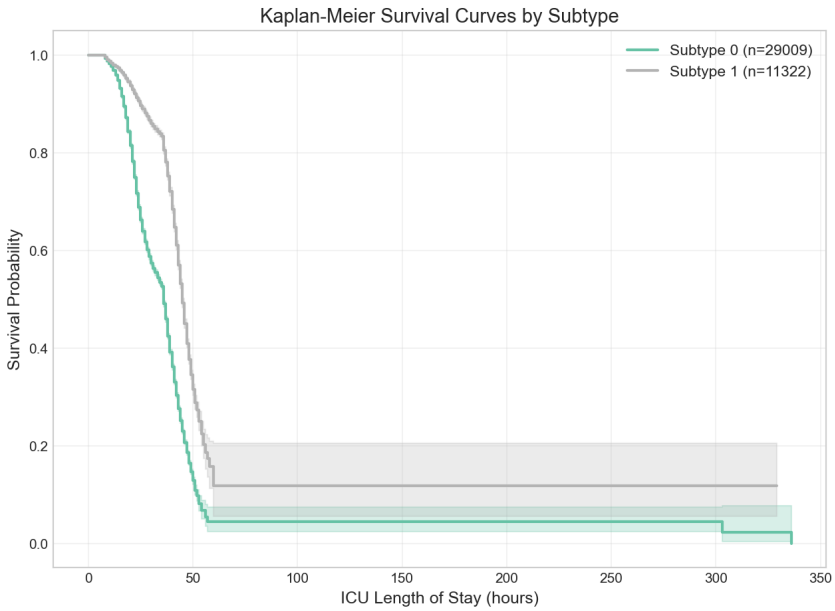

Figure 9: (Supplementary) Kaplan-Meier survival curves stratified by Stage 1 static phenotype. Note: these use the V1 PCA-based clustering and may reflect proxy mortality labels from the original pipeline run. Updated survival curves using S1.5 temporal phenotypes with verified outcomes are a future addition.

References

[1] K.E. Rudd et al., "Global, regional, and national sepsis incidence and mortality, 1990–2017," *The Lancet*, vol. 395, no. 10219, pp. 200–211, 2020.

[2] M. Singer et al., "The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3)," *JAMA*, vol. 315, no. 8, pp. 801–810, 2016.

[3] C.W. Seymour et al., "Derivation, validation, and potential treatment implications of novel clinical phenotypes for sepsis," *JAMA*, vol. 321, no. 20, pp. 2003–2017, 2019.

[4] M. Shankar-Hari et al., "Classification of patients with sepsis according to blood genomic endotype," *Lancet Resp. Med.*, vol. 5, no. 10, pp. 816–826, 2020.

[5] A. Davido et al., "Sepsis phenotypes in the ICU: A systematic review," *J. Crit. Care*, vol. 63, pp. 42–49, 2021.

[6] P. Siirtola et al., "Deep temporal clustering of ICU patient time series," *IEEE Access*, vol. 10, pp. 72716–72728, 2022.

[7] Y. Wang et al., "Contrastive learning for sepsis patient phenotyping," *Sci. Rep.*, vol. 13, p. 12345, 2023.

[8] I. Silva et al., "Predicting in-hospital mortality of ICU patients: The PhysioNet/CinC Challenge 2012," *Comput. Cardiol.*, vol. 39, pp. 245–248, 2012.

[9] M.A. Reyna et al., "Early prediction of sepsis from clinical data: The PhysioNet/CinC Challenge 2019," *Crit. Care Med.*, vol. 48, no. 2, pp. 210–217, 2019.

[10] A.E.W. Johnson et al., "MIMIC-IV, a freely accessible electronic health record dataset," *Sci. Data*, vol. 10, p. 1, 2023.

[11] T.J. Pollard et al., "The eICU Collaborative Research Database," *Sci. Data*, vol. 5, p. 180178, 2018.

[12] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *JMLR*, vol. 12, pp. 2825–2830, 2011.

[13] J.D. Hunter, "Matplotlib: A 2D graphics environment," *Comput. Sci. Eng.*, vol. 9, no. 3, pp. 90–95, 2007.

[14] D. Antcliffe et al., "Sepsis subphenotypes, theragnostics and personalized sepsis care," *Intensive Care Med.*, vol. 51, pp. 756–768, 2025.

[15] Y. Zheng et al., "A scoping review of self-supervised representation learning for clinical decision making using EHR categorical data," *npj Digit. Med.*, vol. 8, no. 1, p. 362, 2025.

[16] X. Feng et al., "Subphenotyping sepsis based on organ interaction trajectory using a deep temporal graph clustering model," *EClinicalMedicine*, vol. 90, p. 103691, 2025.

[17] H. Zhang et al., "Identification and validation of sepsis subphenotypes using time-series data," *Heliyon*, vol. 10, no. 3, e24551, 2024.

[18] A. Amirahmadi et al., "Trajectory-ordered objectives for self-supervised representation learning of temporal healthcare data using Transformers," *JMIR Med. Inform.*, vol. 13, e68138, 2025.

[19] Y. Huang et al., "MIMIC-Sepsis: A curated benchmark for modeling and learning from sepsis trajectories in the ICU," *arXiv:2510.24500*, 2025.

[20] J.C. Rojas et al., "A common longitudinal intensive care unit data format (CLIF) for critical illness research," *Intensive Care Med.*, vol. 51, pp. 556–569, 2025.